Digital Insomnia: Screen Habits & Sleep Analysis

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Why?

The Digital Insomnia: Screen Habits & Sleep dataset peaked my interest because, as a college student, sleep is critical. I often feel that despite going to sleep early and getting 7-8 hours of sleep, I still feel fatigued in the morning and struggle to get out of bed. I know this is a common issue among young adults and adults of all ages.

Primary Question:

How does the timing of screen use influence the relationship between screen time and sleep quality?

Hypothesis:

Using screens for a long period of time has less of an impact when done earlier in the night.

About the Dataset

Digital Insomnia: How Screen Habits Shape Your Sleep take samples from 8500 individuals and observes lifestyles habits that affect sleep like caffeine and exercise, and primarily screen usage. The dataset measures sleep quality using metrics like total sleep hours, REM percentage, deep sleep percentage, and next day fatigue, among others.

user_id	int64
age	int64
gender	object
chronotype	object
occupation	object
bedtime_hour	float64
screen_time_before_bed_min	int64
primary_app_category	object
screen_brightness_pct	int64
blue_light_filter_on	bool
caffeine_intake_mg	int64
caffeine_last_hour	float64
physical_activity_min	int64
room_temperature_f	float64
ambient_noise_db	float64
sleep_onset_latency_min	float64
total_sleep_hours	float64
rem_sleep_pct	float64
deep_sleep_pct	float64
awakenings_count	int64
sleep_efficiency_pct	float64
next_day_fatigue_score	float64
sleep_debt_category	object

Methodology

01 Data Selection

Selected and isolated relevant variables: bedtime, screen time before bed, total sleep hours, REM sleep, deep sleep, and next-day fatigue.

02 Data Grouping

Divided bedtimes into four two-hour groups: 8–10 PM, 10 PM–12 AM, 12–2 AM, and 2–4 AM.

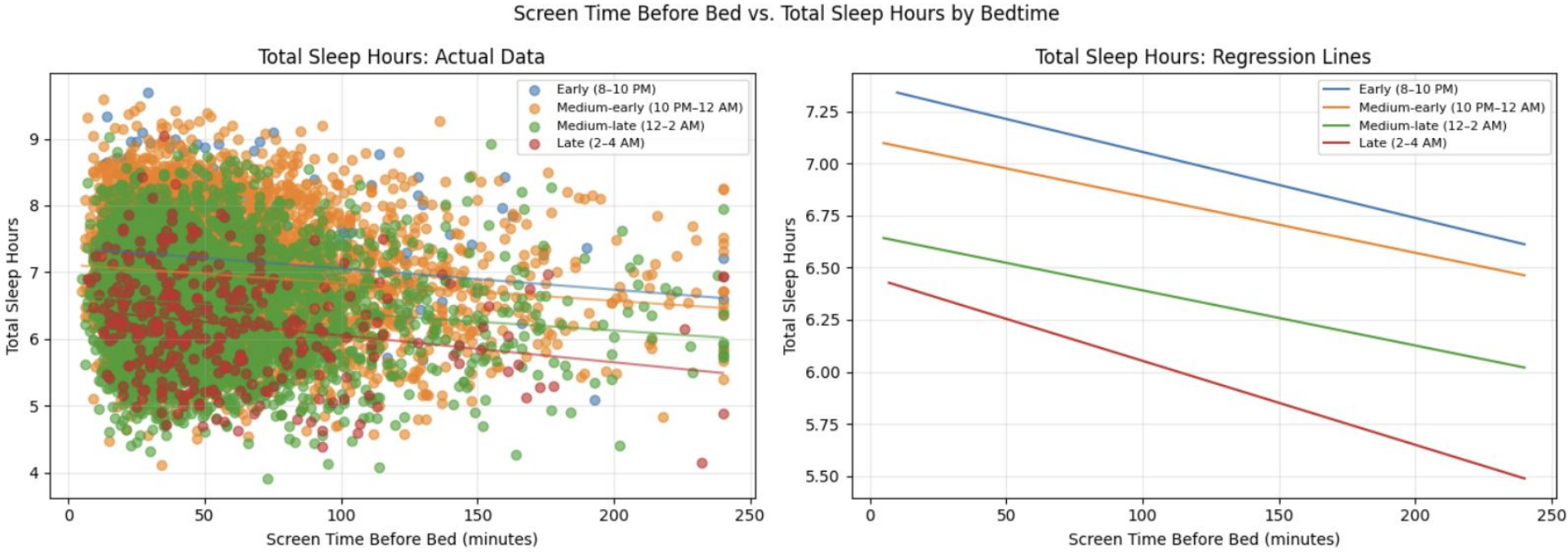
03 Exploratory Analysis

Created scatter plots to examine screen time in relation to total sleep, REM sleep, deep sleep, and next-day fatigue for each bedtime group.

04 Trend Analysis

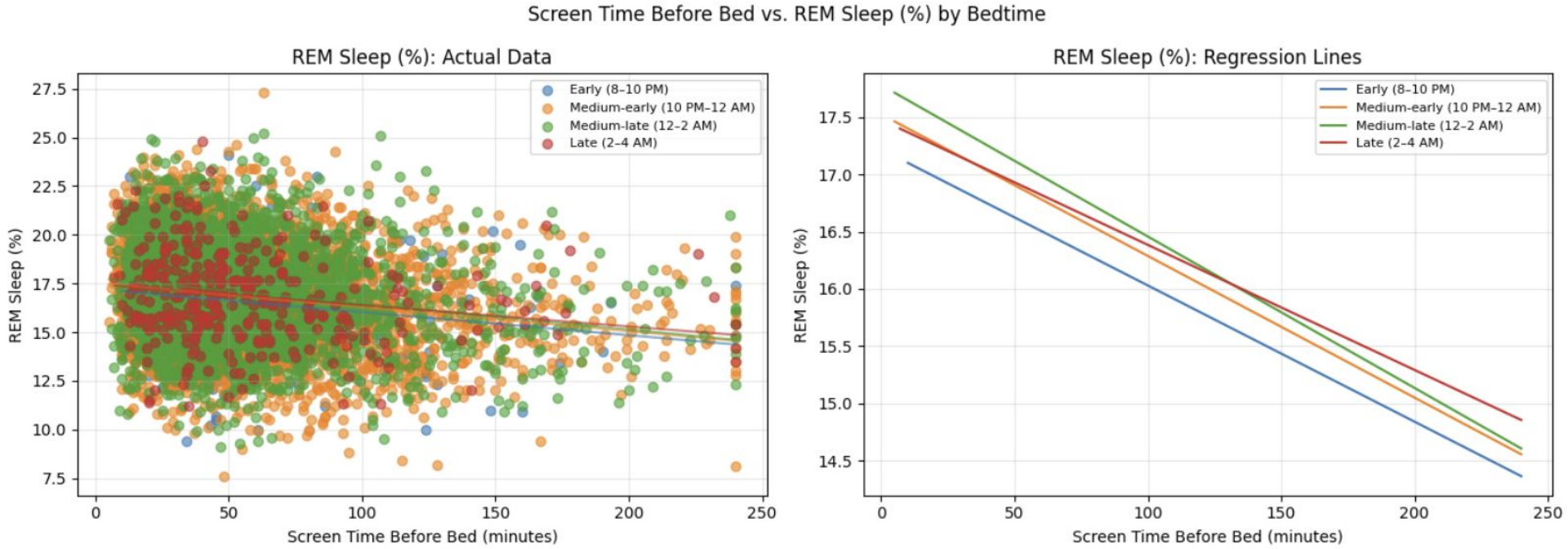
Added separate regression lines for each bedtime group to identify trends. Compared the direction and strength of the relationships to investigate whether earlier screen use is associated with less impact on sleep.

Graph 1: Screen Time Vs Total Sleep



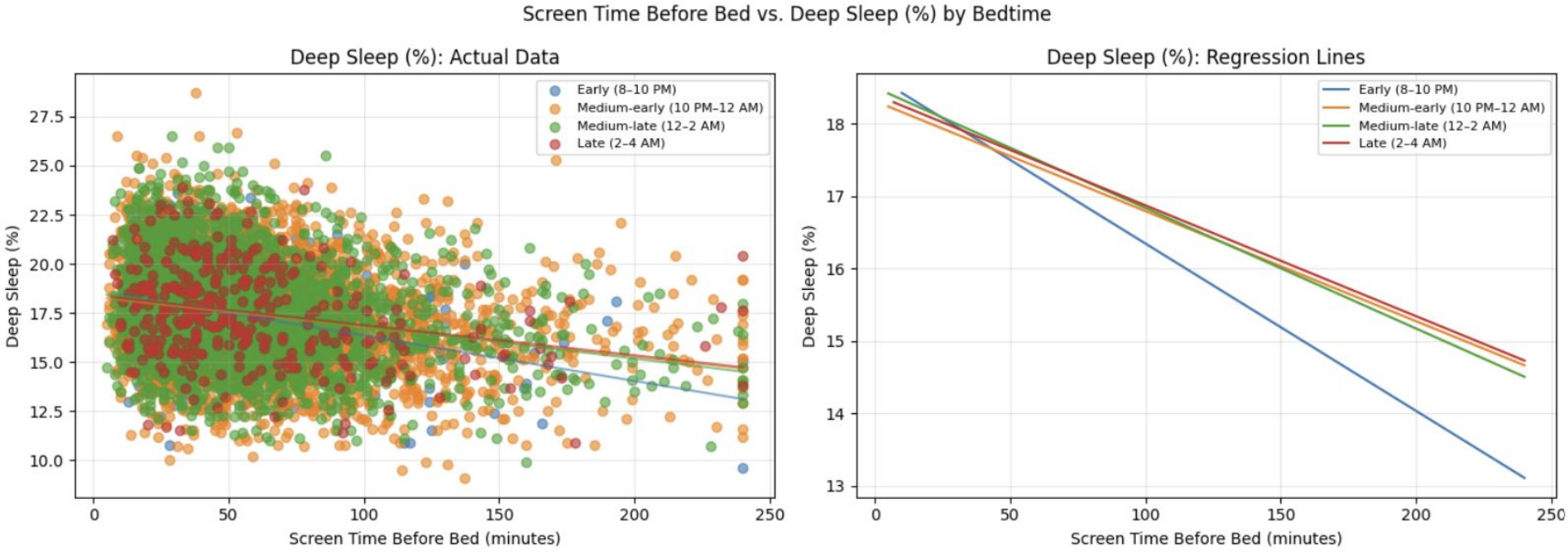
*Note that the y-axis scale is different on each graph

Graph 2: Screen Time Vs REM %



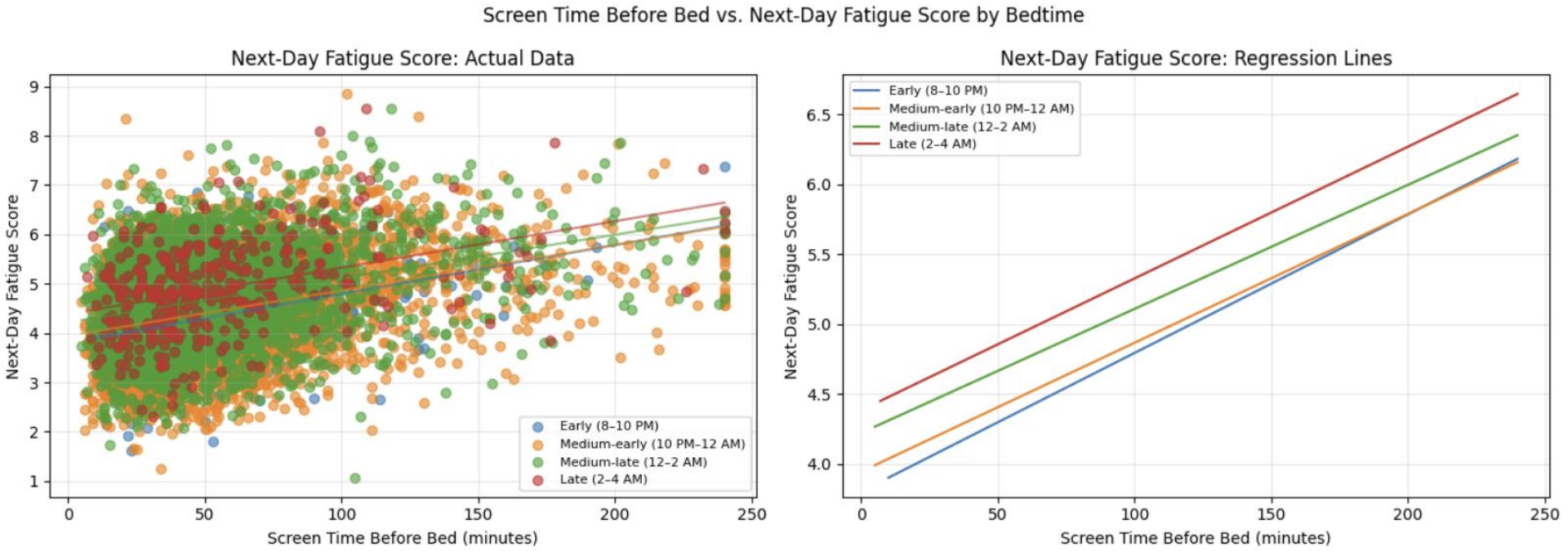
*Note that the y-axis scale is different on each graph

Graph 3: Screen Time Vs Deep Sleep %



*Note that the y-axis scale is different on each graph

Graph 4: Screen Time Vs Next-Day Fatigue



*Note that the y-axis scale is different on each graph

Key Findings

Screen Time and Sleep Quality

Screen time was negatively associated with sleep quality across the metrics analyzed, including total sleep hours, REM sleep, and deep sleep. More screen time was also associated with greater next-day fatigue.

Bedtime Makes Little Difference

The relationships between screen time and most sleep metrics were relatively similar across the four bedtime groups.

An Exception: Deep Sleep

The early bedtime group had a steeper regression line for deep sleep, suggesting a stronger relationship between screen time and deep sleep in this group.

Conclusion

Our hypothesis was disproven. We expected screen time to have less of an impact on sleep when it happened earlier in the night, but the relationship between screen time and most sleep metrics was pretty similar across all four bedtime groups.

I was surprised by how little the timing of screen use seemed to matter. The biggest exception was deep sleep, where the early bedtime group had a steeper regression line than the other groups. This was unexpected and made me wonder why deep sleep showed a different pattern.

There were a few limitations to our analysis. We only looked at certain variables, leaving out others in the dataset, such as caffeine intake, screen brightness, and exercise. Also, grouping bedtimes into two-hour intervals may have hidden smaller differences. Most importantly, our results only show correlation, not causation, so we cannot say that screen time or bedtime directly affects sleep.

Overall, even though our hypothesis was disproven, the results were interesting and showed me that sleep quality may depend on more factors than just screen time and bedtime. Looking at the other variables could help us better understand these relationships.

Additional Research Questions for this Dataset

1. Does Age Influence the Relationship Between Screen Time and Sleep?

- Research question: Does screen use before bed have a greater impact on sleep quality for younger people than older people?

- Younger and older people may have different screen habits and sleep patterns. I want to see if age helps explain why screen time affects people differently.

2. Does the Type of Screen Content Matter?

- Research question: Is short-form content, like TikTok and Instagram Reels, more strongly associated with poorer sleep than longer-form content, like watching a TV show?

- Short-form content seems especially addictive and can make it easy to lose track of time. I want to explore whether the type of content people consume matters more than simply how long they spend on their screens.